

DATA STRUCTURES AND ALGORITHMS
PREVIOUS YEAR QUESTION PAPER

1. Explain the process of insertion and deletion in a singly linked list with example.
2. What is a doubly linked list? Explain node structure.
3. Explain push and pop operations in a stack with an example.
4. Write an algorithm to check balanced parentheses using stack.
5. Explain enqueue and dequeue operations in a queue with diagram.
6. Differentiate between stack and queue.
7. What is a binary tree? Explain preorder, inorder and postorder traversal.
8. Construct a binary search tree (BST) for the given elements and explain insertion.
9. Explain BFS and DFS traversal techniques in a graph with example.
10. What is Dijkstra's algorithm? Explain shortest path finding in a weighted graph.
11. Explain bubble sort algorithm with time complexity analysis.
12. Write and explain merge sort algorithm with example.
13. Explain quick sort algorithm and its time complexity.
14. What is binary search? Explain with an example and time complexity.
15. Explain linear search algorithm and compare it with binary search.
16. Explain the concept of a node in a linked list and how deletion works.
17. What is a tree traversal? Explain BST operations.
18. Explain graph representation using adjacency list and adjacency matrix.
19. Compare bubble sort, merge sort and quick sort based on time complexity.
20. Explain stack applications with real-world examples.